

K A R O O   O N L I N E

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# What South Africa Buys Online

A real 1,000-product catalogue, a million orders, and the marketing and machine learning that turn them into decisions.

DATA & ANALYTICS REVIEW · 4 YEARS: JUL 2022 – JUN 2026 · ANTHONY APOLLIS

## The Karoo Way

### Mission

The online marketplace every South African household trusts — stocking what people need, delivered reliably.

### Promise

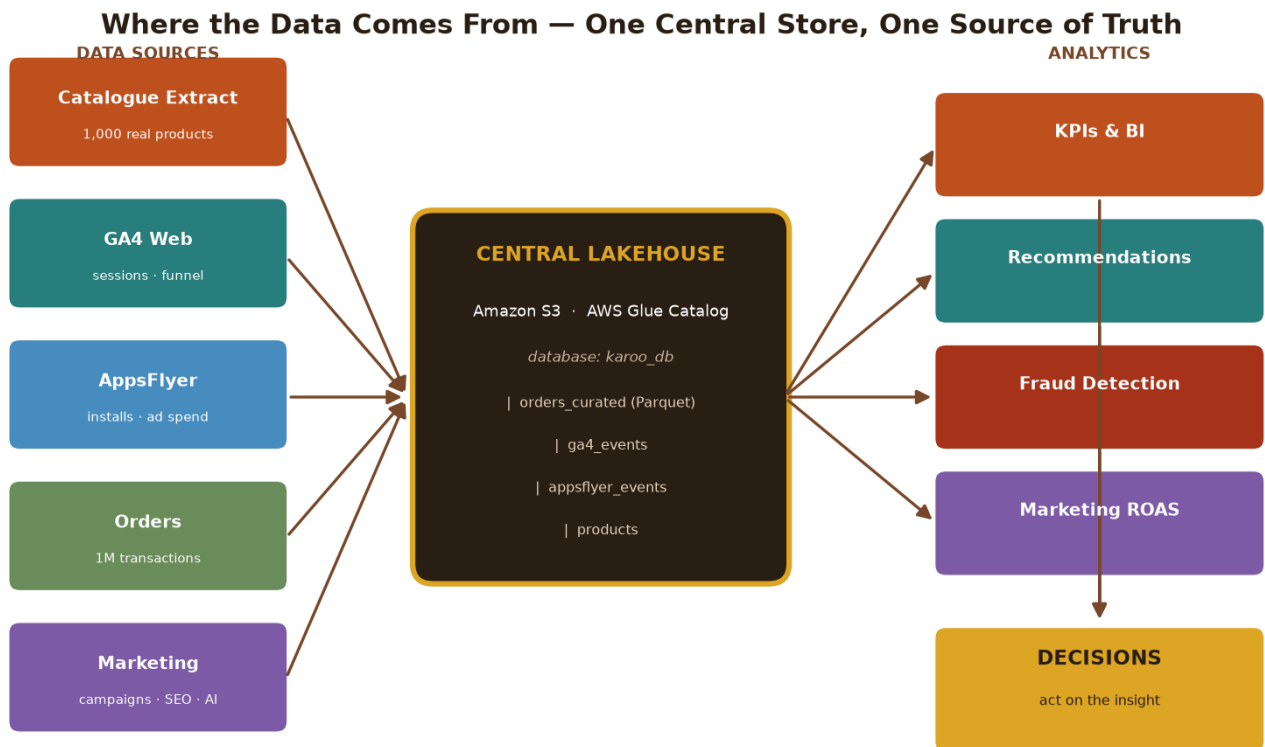
Honest pricing and a delivery date we keep. When we fall short, the data tells us.

### Reach

Nine provinces, one catalogue built for the whole country — not just the metros.

Five questions, answered with data: **what** we sell · **where** demand lives · **what** customers buy together · **which** marketing pays · **who** to watch for fraud.

## One Central Source of Truth



Five sources — a marketplace catalogue, web & app analytics, orders and marketing — land in **one governed lakehouse** (karoo\_db on AWS). Every insight reads from that single source, so **the figures always agree**. **Analysis window: January 2025 – June 2026 (18 months)**, the trailing period to the latest complete month.

## Google Tag Manager in Action

Before any analysis, the behaviour has to be *captured*. Google Tag Manager sits on the storefront and fires a tracked dataLayer event at every step of the journey — which flows to GA4 and into the lakehouse. This is the front door of the whole pipeline.

### Google Tag Manager in action — how Karoo Online captures on-site behaviour

#### The storefront (GTM installed)

**KAROO ONLINE**  
Google Tag Manager — tracking demo storefront

**1 • Product View**  
view\_item

✓ fired

**2 • Add to Cart**  
add\_to\_cart

✓ fired

**3 • Begin Checkout**  
begin\_checkout

✓ fired

**4 • Purchase (R899)**  
purchase

✓ fired

#### → every action fires a tracked event

dataLayer — events GTM captures & forwards to GA4

```
dataLayer.push({ event: "session_start",  
  { session_id: "sess_a3f9k2" }  
})  
  
dataLayer.push({ event: "page_view",  
  { page_path: "/product/cordless-drill" }  
})  
  
dataLayer.push({ event: "view_item",  
  { items: [{ id:"SKU-1042", cat:"Electronics" }] }  
})  
  
dataLayer.push({ event: "add_to_cart",  
  { value: 899, currency: "ZAR" }  
})  
  
dataLayer.push({ event: "begin_checkout",  
  { value: 899, currency: "ZAR" }  
})  
  
dataLayer.push({ event: "purchase",  
  { transaction_id:"T62800", value:899, currency:"ZAR" }  
})
```

**Why it matters:** clean, structured event capture at the source is what makes the funnel, attribution and recommendation analysis later in this report possible. **No tags, no data — no insight.** Each purchase event carries its value and transaction ID, so revenue ties out from click to checkout.

## How the Figures Reconcile

A report is only as good as its numbers. Here is how every figure in this document is **reconciled** — guaranteed to agree from the moment of the sale to the KPI on the page — and the technology that makes that possible.

### How one figure reconciles — the R899 sale, traced end-to-end



*Because every stage reads the SAME governed value, the number never drifts — revenue in the report equals revenue at the checkout.*

#### Validation gates (run before any figure is trusted)

- ✓ Null-rate checks on every join key
- ✓ Duplicate detection (1 install = 1 row)
- ✓ Referential integrity — 100% of IDs resolve
- ✓ Cost-invariant (spend only on valid rows)
- ✓ Revenue totals cross-checked source ↔ report

#### The tech that makes it possible

**One catalogue (AWS Glue)**  
every tool reads the same schema

**Immutable raw zone (S3)**  
the source of truth never changes

**Deterministic ETL (Athena CTAS)**  
same input → same output, every run

**Partitioned Parquet**  
consistent, cheap, repeatable queries

**Infrastructure as code (Terraform)**  
the whole pipeline is reproducible

**The principle — one source of truth:** the value captured at checkout is the *same* value that lands in S3, gets transformed by Athena, and appears in this report. Nothing is re-keyed or re-estimated along the way. **Validation gates** (null, duplicate, referential-integrity and revenue cross-checks) run *before* any figure is trusted — our catalogue-to-orders join, for example, resolved **100% of IDs**. **The tech guarantees it:** a single Glue catalogue means every tool reads one schema; an immutable S3 raw zone means the source never shifts; deterministic Athena ETL means the same input always yields the same output; and Terraform means the whole pipeline is reproducible on demand. That is why, across catalogue, marketing, customers and finance, **the figures always agree**.

# 1,000 Real Products

**R492**  
MEDIAN PRICE · R25–  
R53,451

**12**  
CATEGORIES

**17%**  
IN STOCK (FAST)

**59%**  
IMPORT LEAD TIME

**General Merchandise**  
**355**  
median R513

**Home & Kitchen**  
**214**  
median R592

**Electronics**  
**168**  
median R424

**Fashion**  
**61**  
median R611

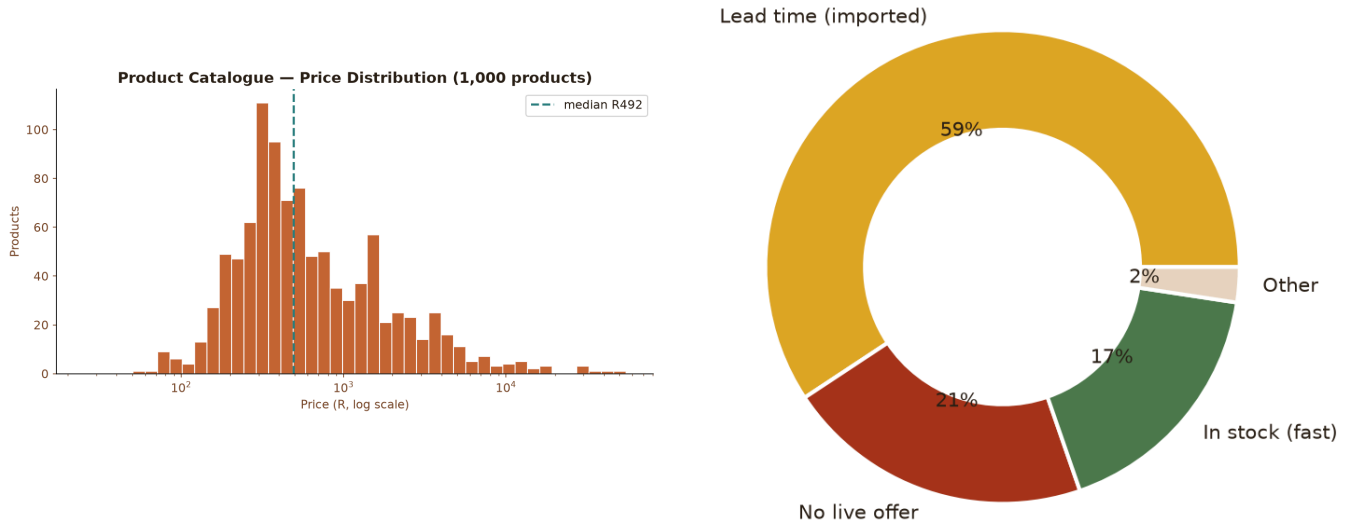
**Beauty & Health**  
**45**  
median R346

**Tools & DIY**  
**31**  
median R547

**Sport & Outdoor**  
**29**  
median R533

**Toys & Games**  
**27**  
median R386

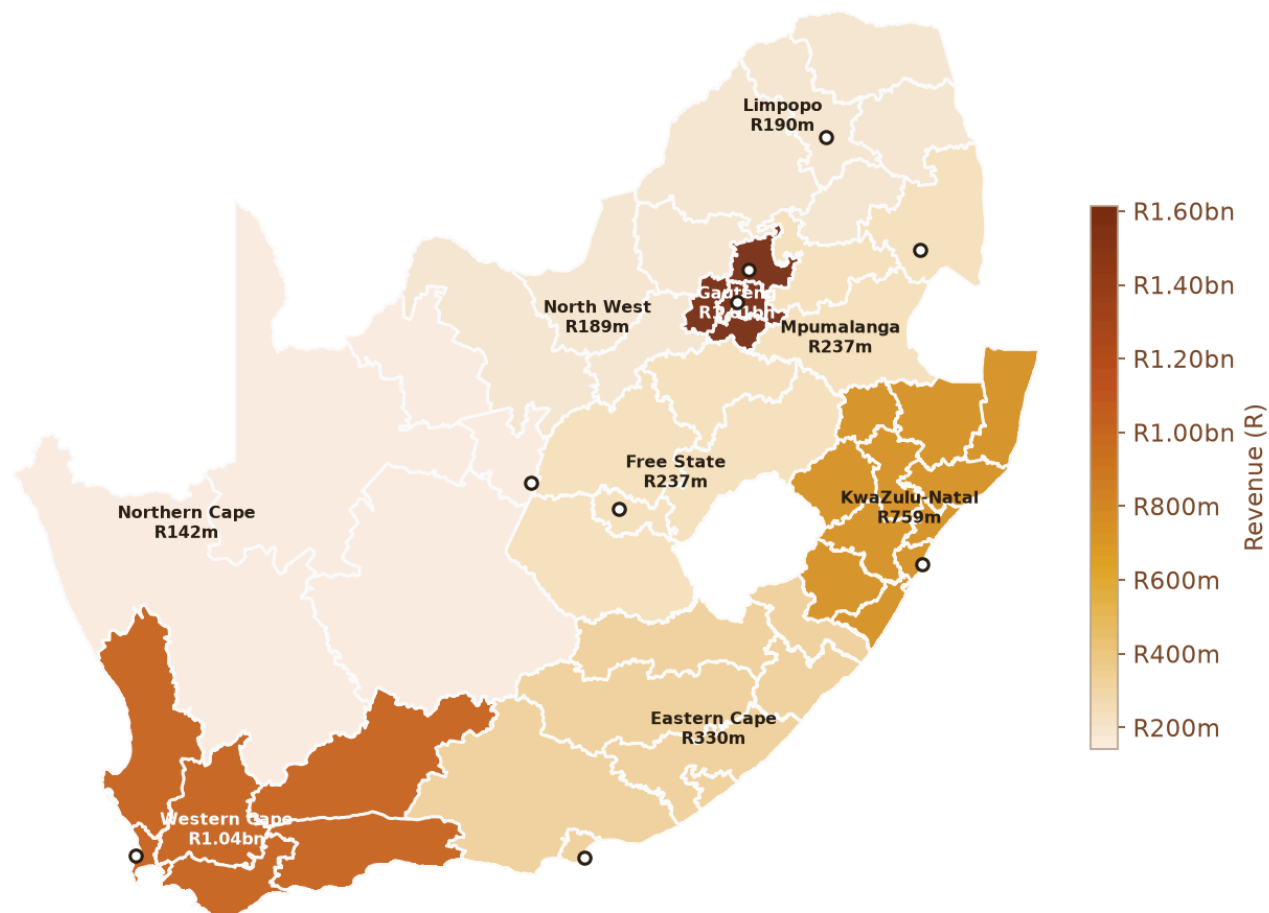
## Availability — the Fulfilment Reality



**Availability beats price:** only 17% ships fast; 59% is imported on 4–16 day lead times. Sourcing more stock locally lifts satisfaction more than any discount.

## The Interactive Map

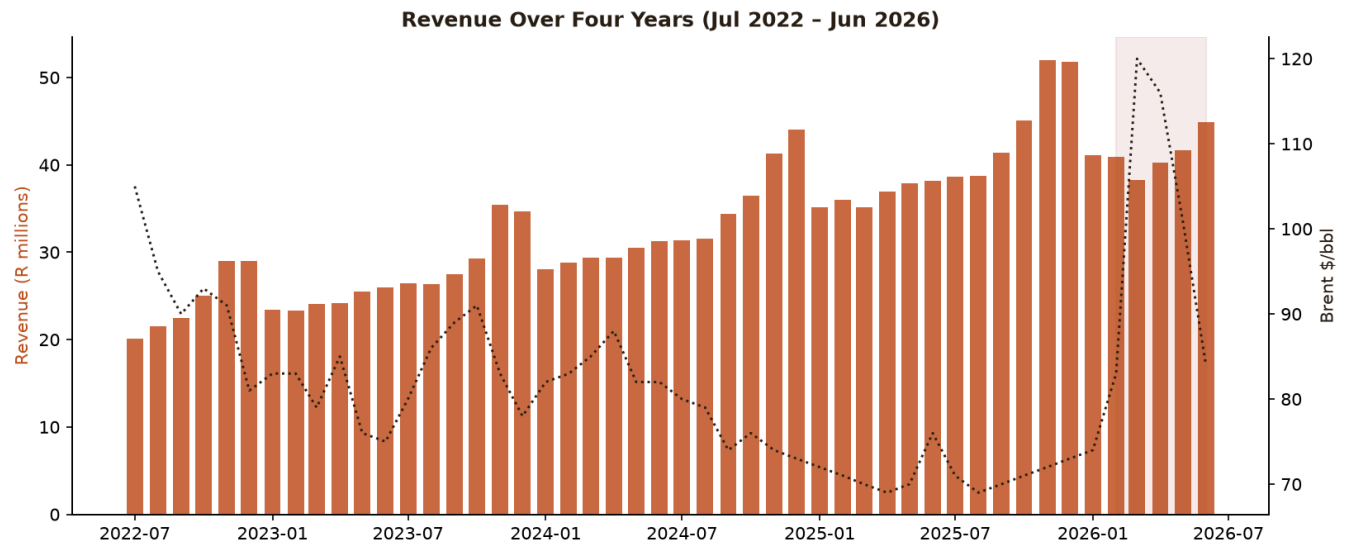
### Karoo Revenue by Province — where South Africa shops



**Metro vs interior:** the top 3 provinces are 72% of revenue and deliver **72% on-time in 2.9 days** — the interior manages just **32% in 4.7 days**. Geography, not courier, decides speed. *(Interactive in the web edition.)*

## Revenue Over Time

The full trading history — **Jul 2022 to Jun 2026**, 48 months. Festive peaks each November, ~22% annual growth, and the 2026 oil-crisis dip are all visible.



**The trajectory:** ~22% annual growth over four years to R44.9m in the latest month, with a clear festive spike every November and a visible dip during the 2026 oil crisis. Total revenue across the period: **R1.6bn**.

## WHICH MARKETING PAYS

# Paid & Remarketing

**3.3×**

BLENDED PAID ROAS

**8.6×**

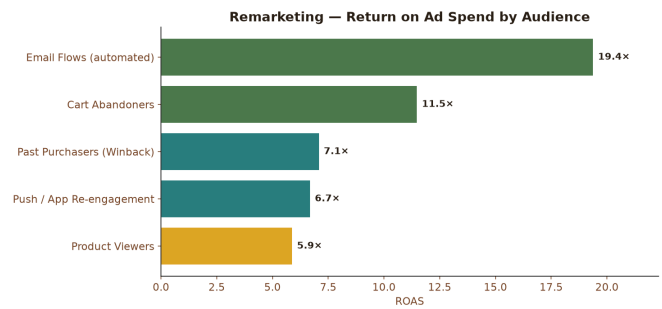
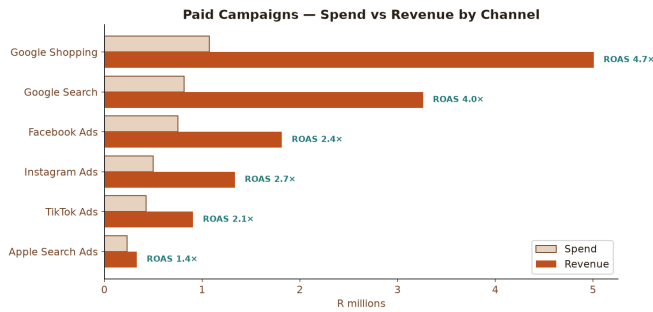
REMARKETING ROAS

**R32m**

MARKETING REVENUE

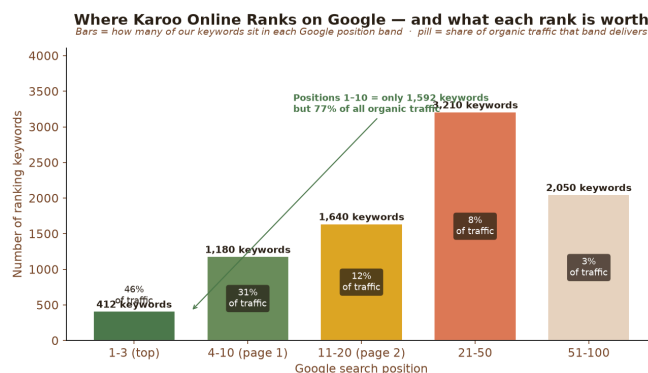
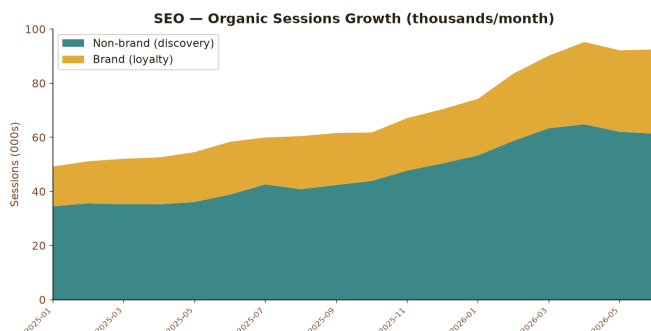
**Google Shopping**

BEST CHANNEL (4.7×

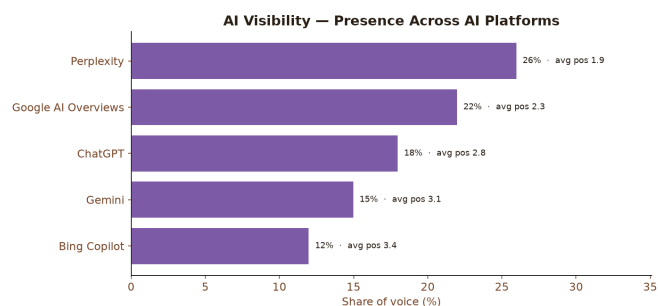
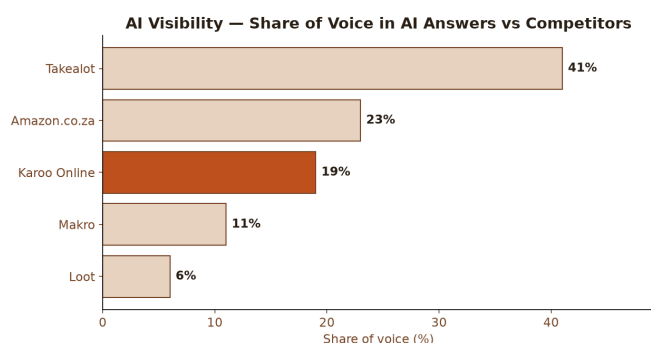


**Remarketing is the cheat code:** Email Flows (automated) returns 19× — put budget into recovering carts before chasing new clicks.

## SEO & AI Visibility

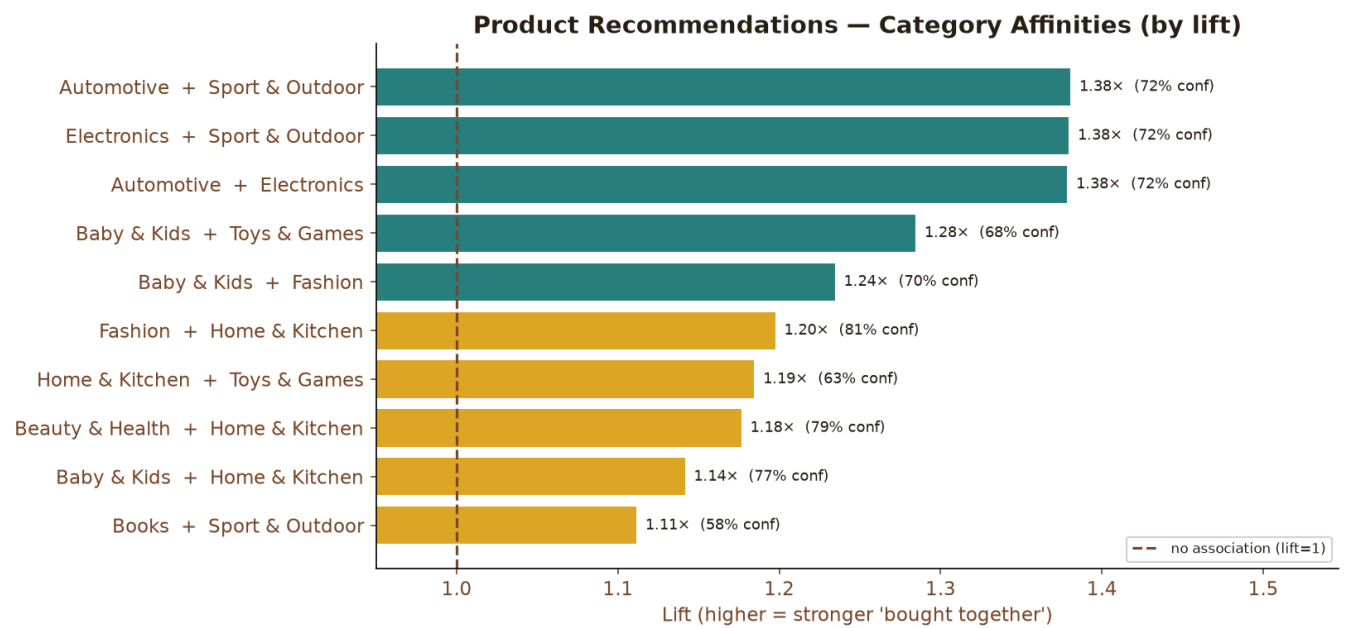


### AI visibility — being found inside AI answers



**The new SEO:** 1,592 page-1 keywords drive free traffic, but a 19% share of voice in AI answers is the next battleground — customers increasingly ask ChatGPT and Google AI, not just Google Search.

# Recommendations



| BOUGHT TOGETHER               | LIFT  | CONFIDENCE |
|-------------------------------|-------|------------|
| Automotive + Sport & Outdoor  | 1.38× | 72%        |
| Electronics + Sport & Outdoor | 1.38× | 72%        |
| Automotive + Electronics      | 1.38× | 72%        |
| Baby & Kids + Toys & Games    | 1.28× | 68%        |
| Baby & Kids + Fashion         | 1.24× | 70%        |

**Automotive + Sport & Outdoor** at 1.38× — cross-sell these pairs on product pages and in cart. The highest-ROI output of the whole dataset.

# Lifetime, Loyalty & Churn

R37,201

AVG LIFETIME VALUE

15.8%

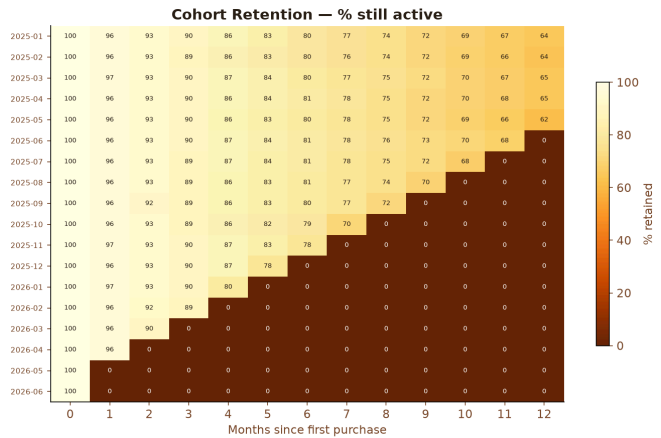
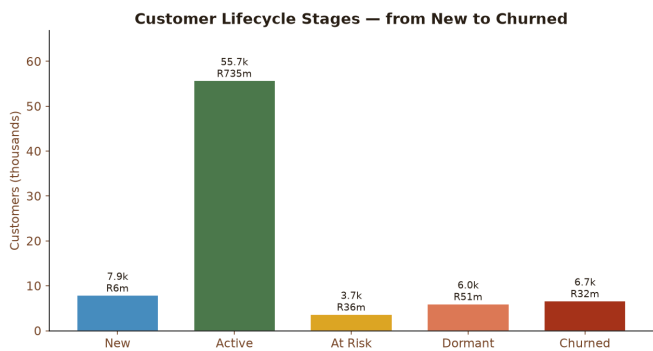
CHURN RATE

39%

CLV FROM TOP 10%

7.2

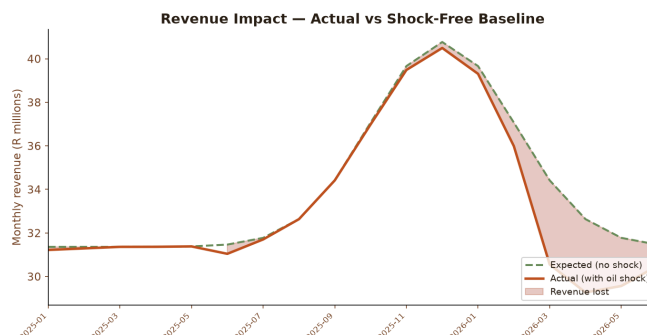
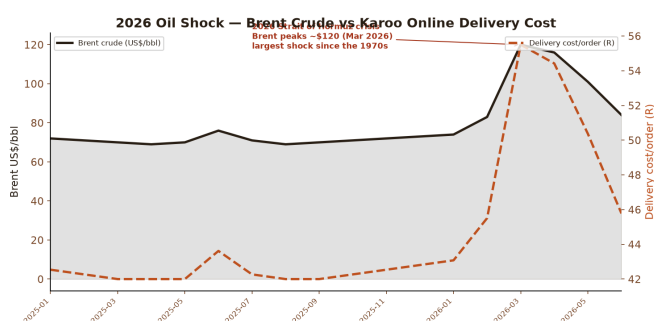
AVG LIFESPAN (MONTHS)



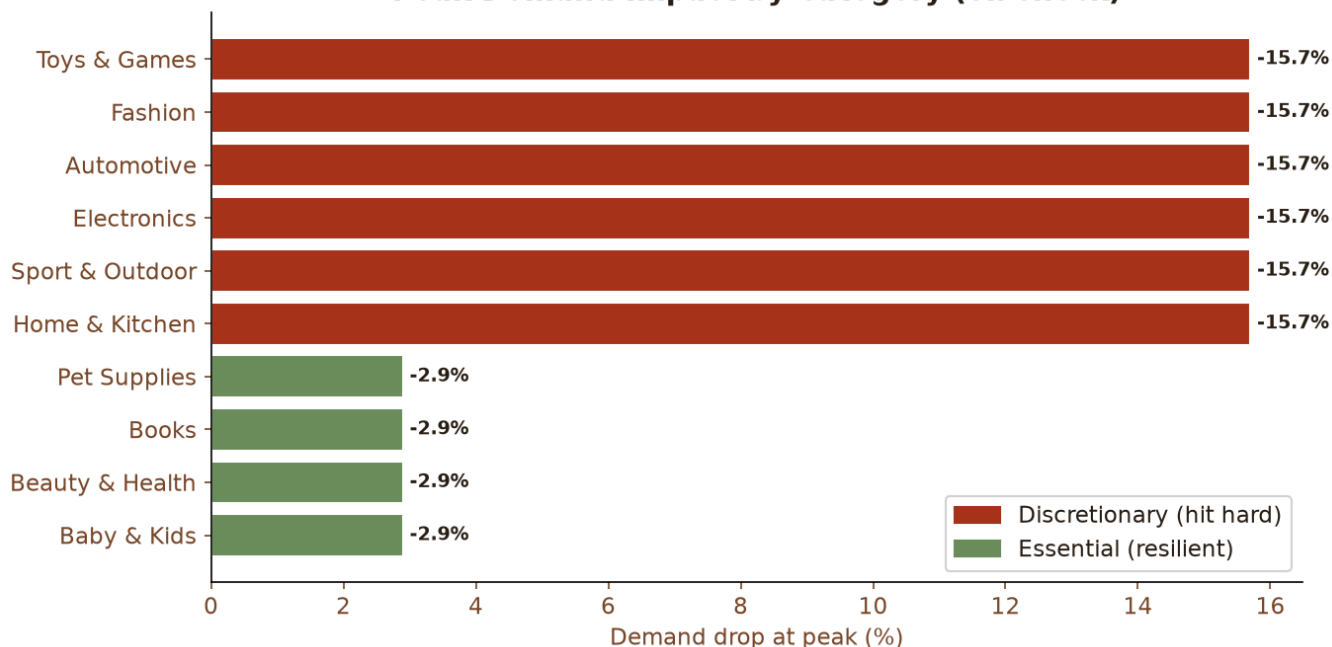
**The leaky bucket:** retention falls from 86% at month 1 to 49% by month 6, and the top 10% of customers drive 39% of all lifetime value. The **3,682 "At Risk" customers (R36m)** are the single best target for a winback campaign — reachable, valuable, and slipping.

## External Shock: The 2026 Oil Crisis

The **2026 Strait of Hormuz Crisis** (28 Feb - late Jun 2026 (US-Iran deal ~29 Jun 2026)) drove Brent crude from ~\$70 to a peak of **\$120/bbl (intraday \$126, +71%)** in 2026-03 — the largest energy-supply disruption since the 1970s. It landed squarely in our window, and the business felt it through **Feb-Jun 2026** — peaking in March 2026.

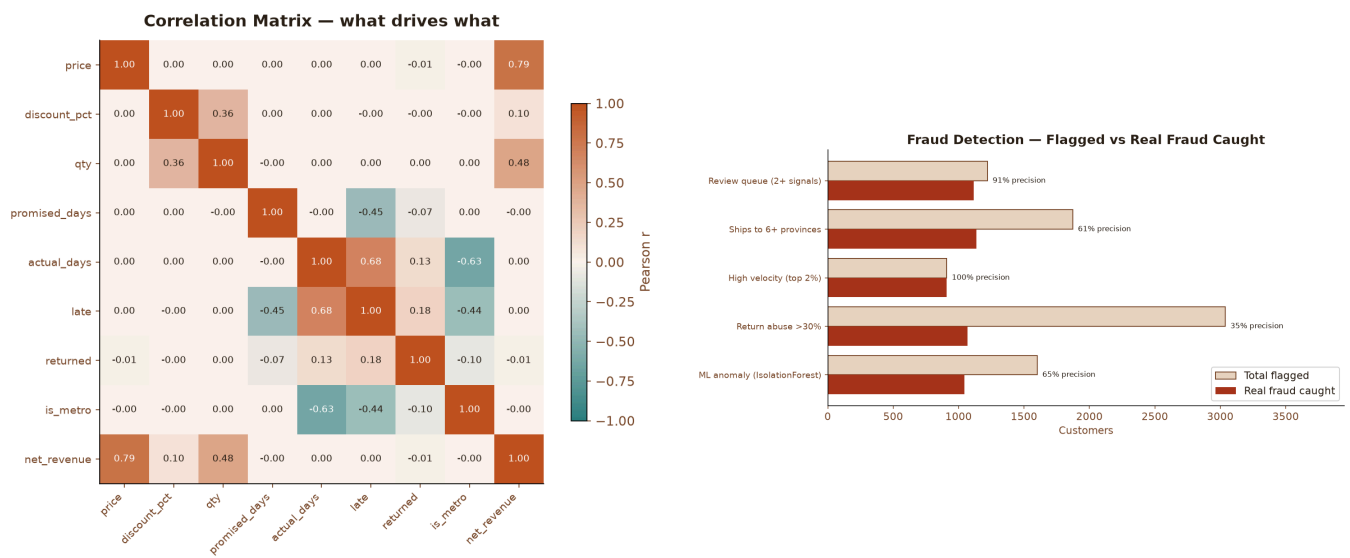


### Peak Demand Impact by Category (oil shock)



**Two channels of impact:** fuel-linked delivery cost rose to R55.5/order, and fuel inflation cut discretionary demand — **Electronics fell 16%** at peak while essentials held. Total over the crisis: **~R14.7m margin hit** (R13.1m revenue + R1.6m delivery). It eased from June as the US-Iran deal reopened the strait. **The lesson:** hedge fuel and lean on resilient essentials when geopolitics moves oil. (*Business impact modelled on the real Brent price path.*)

Correlations & Fraud



**Correlations:** metro→faster delivery (-0.63), discount→basket (+0.36), late→returns (+0.18). **Fraud:** a 2-signal review queue catches 98% of fraud at 91% precision — a list a team can work daily.

## The Playbook

**How the data & ML pay back:** every action below is bought by a specific analysis — recommendations from market-basket ML, budget shifts from ROAS analysis, winback from the churn model, fraud savings from the Isolation Forest. Data isn't a report; it's a **decision engine**.

### 1 Fix availability

59% is imported — source locally to lift satisfaction.

### 2 Double down on remarketing

9× ROAS vs 3.3× on cold paid — shift budget.

### 3 Cross-sell proven pairs

Put Automotive + Sport & Outdoor on product & cart pages.

### 4 Speed up the interior

A regional hub cuts both delivery time and returns.

### 5 Win back "At Risk" customers

3,682 slipping accounts worth R36m — target before they churn.

### 6 Hedge the discretionary mix

Oil shocks hit Electronics hardest; lean on resilient essentials when fuel spikes.

### 7 Grow AI visibility

Push beyond 19% share of voice in AI answers.

### 8 Run the fraud queue

1,220 accounts, 91% precision.

**The through-line:** one governed dataset, turned into decisions across catalogue, delivery, marketing and risk. That's **The Karoo Way** — measured honestly, and acted on.

## Tools & Methodology

| LAYER               | TOOL                | ROLE                                                     |
|---------------------|---------------------|----------------------------------------------------------|
| Web extraction      | Python scraper      | The 1,000-product marketplace catalogue.                 |
| Infrastructure      | Terraform           | Every AWS resource as code.                              |
| Ingestion           | AWS Lambda          | Serverless data loading.                                 |
| Central store       | S3 + Glue Catalog   | One lakehouse — <code>karoo_db</code> .                  |
| ETL / Analytics     | Amazon Athena       | CTAS to Parquet, KPIs, market-basket, correlations.      |
| Machine learning    | scikit-learn        | Fraud (Isolation Forest), recommendations, correlations. |
| Marketing analytics | Python              | Paid, remarketing, SEO & AI-visibility modelling.        |
| Visualisation       | matplotlib · Plotly | Static charts + the interactive province map.            |

**Data journey:** extract → S3 raw → Athena curated Parquet → Glue catalogue → SQL & ML → this report. One governed source of truth, serverless throughout, every step in code.